



STRUCTURAL DRIVERS OF PER-CAPITA CO₂ EMISSIONS: A CROSS-COUNTRY ANALYSIS (1990–2021)

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INTRODUCTION

1.1 MOTIVATION AND CONTEXT

The inclusive aim of this study is to identify and understand the structural drivers of per-capita CO₂ emissions across countries over time. Rather than focusing exclusively on the technical aspects of energy production and consumption, the analysis integrates economic, social, and demographic dimensions to capture the broader determinants of emission trajectories. By doing so, the study seeks to provide a comprehensive perspective on how national energy systems interact with development pathways and societal structures in shaping carbon footprints.

The project is explicitly designed at the per-capita level, in order to evaluate not only the aggregate emissions of countries but also the relative emissions attributable to individual citizens. This perspective highlights inequalities in carbon responsibility, where the average citizen in some countries emits an order of magnitude more than in others, despite similarities in access to modern energy or economic development levels.

1.2 RESEARCH QUESTIONS AND OBJECTIVES

The research is structured around four main questions (RQs):

- **RQ1 – Drivers of CO₂ emissions:** Which socio-economic and energy variables explain cross-country and temporal variation in CO₂ emissions per capita?
- **RQ2 – Predictive capacity:** How well can machine learning models (Random Forest, XGBoost) predict per-capita CO₂ emissions?
- **RQ3 – Country deviations (“fair residuals”):** Which countries emit more or less CO₂ than expected once structural drivers are accounted for?
- **RQ4 – Clustering of energy profiles:** Can countries be grouped into distinct clusters based on their energy mix and efficiency?

1.3 CONTRIBUTION AND STRUCTURE

To address these questions, the study employs a multi-method strategy. Mixed-effects regression models are used to capture both cross-country variations, while accounting for unobserved heterogeneity through country-specific intercepts. Autoregressive structures are incorporated to deal with serial correlation in the error terms. Machine learning methods, specifically Random Forests and XGBoost, complement the analysis by providing a resilient predictive benchmark and by highlighting non-linear relationships between variables. Finally, Principal Component Analysis combined with K-Means clustering is used to reduce dimensionality and to classify countries into structurally meaningful groups, thereby uncovering systematic differences in energy profiles and associated emission levels.

DATA AND VARIABLE

2.1 DATA SOURCE

This analysis relies on the World Development Indicators (WDI), the flagship statistical database of the World Bank. The WDI is widely recognized as a reliable source for monitoring global economic, social, and environmental development, drawing on harmonized submissions from national statistical offices and international agencies. Each indicator undergoes methodological review and consistency checks by World Bank staff, ensuring cross-country comparability and temporal consistency.

2.2 DATA OVERVIEW

For the purposes of this study, we constructed a balanced panel covering the years 1990–2021 for over 100 countries worldwide. The dataset combines more than twenty indicators capturing the structure of energy systems, socio-economic development, and demographic context. At its core lies per-capita CO₂ emissions (metric tons per capita), which serves as the dependent variable throughout the analysis. Complementary indicators include:

- **Energy system variables:** electricity consumption per capita, energy use per unit of GDP, fossil fuel shares (coal, oil, gas), renewable and nuclear shares, energy imports, and transmission losses.
- **Socio-economic variables:** GDP per capita, sectoral composition (agriculture, industry, services), and trade openness.
- **Demographic indicators:** urbanization rates and electricity access.

This combination of variables allows the study to go beyond a narrow focus on energy supply, by incorporating the economic and social dimensions that shape national emission patterns. The broad temporal coverage also enables the investigation of both structural cross-country differences and dynamic within-country trends.

2.3 VARIABLE DESCRIPTION

The dataset combines **environmental, energy, economic, and demographic indicators** to capture the multiple dimensions shaping cross-country variation in per-capita CO₂ emissions. The panel structure allows the analysis to jointly account for structural differences across countries and temporal evolution within countries.

2.3.1 Target Variable: CO₂ Emissions per Capita

Per-capita CO₂ emissions (metric tons per person), excluding emissions from land-use, land-use change, and forestry (LULUCF).

This indicator represents the average carbon footprint attributable to a citizen within each country and serves as the outcome variable for all subsequent econometric and machine-learning analyses.

2.3.2 Energy system indicators

- **Electricity consumption per capita** and **total energy use per capita**, capturing the overall scale and intensity of energy demand within a country.
- **Energy efficiency**, measured as GDP per unit of energy use, reflecting how effectively economic output is generated from energy inputs.
- **Electricity generation mix**, expressed through the shares of coal, oil, gas, nuclear, and renewable sources, providing direct information on the carbon intensity of electricity production.
- **Structural energy characteristics**, including net energy imports and transmission and distribution losses, capturing dependence on external energy sources and inefficiencies within national energy systems.

2.3.3 Socio-economic indicators

- **GDP per capita** (constant international dollars), proxying the level of economic development and purchasing power.
- **Sectoral composition of value added** (industry and services), reflecting structural differences in production systems and associated energy requirements.
- **Trade-related indicators**, capturing integration into global markets and exposure to external energy and price shocks.

2.3.4 Demographic and social indicators

- **Urban population share**, representing the spatial concentration of energy demand and differences in infrastructure, transport systems, and household energy use.

- **Access to electricity**, measuring the degree of electrification and inclusiveness of national energy systems, particularly relevant for cross-country comparisons at different development stages.

2.3.5 Summary of Variable Framework

Taken together, this set of indicators enables a **multi-dimensional analysis** of the structural and contextual determinants of per-capita CO₂ emissions, moving beyond a narrow focus on production to incorporate energy systems, economic structure, and demographic context.

2.4 OPERATIONALIZATION AND EMPIRICAL SPECIFICATION OF VARIABLES

The analysis is based on a structured set of energy, economic, and demographic indicators drawn from the World Bank's World Development Indicators. A detailed overview of all selected variables, including definitions and measurement units, is reported in Appendix (Table 1). For the purposes of the empirical work, only a subset of variables was retained in the analysis, namely those considered the most representative, non-redundant with respect to alternative proxies, and most relevant for capturing structural drivers of CO₂ emissions. Here we provide a description of the target variable and the main covariates included in the models.

2.4.1 Target Variable Construction

The dependent variable is CO₂ emissions per capita, measured in metric tons per person and excluding emissions from land-use, land-use change, and forestry (LULUCF). This indicator captures the average carbon footprint attributable to a citizen in each country, thus providing a normalized basis for cross-country comparison.

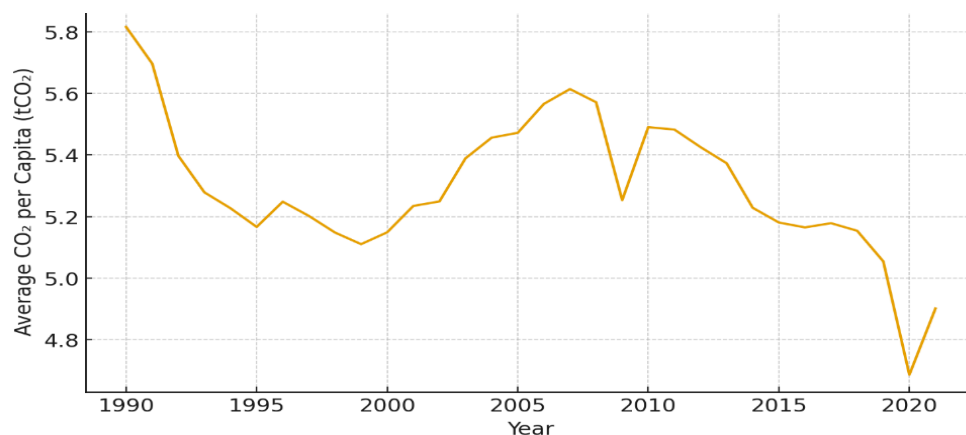


FIGURE 1- GLOBAL AVERAGE TREND OF PER CAPITA CO₂ EMISSIONS

Figure 1 reports the evolution of the global average of per-capita CO₂ emissions between 1990 and 2021. The series fluctuates around a relatively stable mean of 5-6 tons per capita, with modest declines after the mid-2010s and a visible contraction in 2020, coinciding with the global slowdown caused by the COVID-19 pandemic. This overall stability masks sharp contrasts across countries: while many developing economies remain well below the global average, fossil-fuel-intensive economies register values several times higher.

2.4.2 Transformation and Scaling of Covariates

The dataset initially comprised twenty-one indicators from the World Bank's *World Development Indicators* (WDI), covering energy, economic, and demographic dimensions for the period 1990–2021. From this pool, only the most representative and analytically relevant variables were retained, based on their theoretical significance and empirical coherence in explaining per-capita CO₂ emissions.

- **Energy-related indicators** capture both the scale and composition of national systems. Electricity consumption per capita (kWh/person) and total energy use per capita (kgoe/person) measure energy demand, while GDP per unit of energy use (PPP-adjusted constant US\$/kgoe) reflects efficiency.

- **The energy mix** is represented by the shares of coal, gas, oil, nuclear, hydro, and other renewables in total electricity generation, all expressed as percentages to ensure comparability across countries. Renewable energy consumption (as a share of final energy use) and net energy imports (as a share of total energy use) complement this framework by indicating decarbonization and dependency levels.
- **The economic structure** is represented by GDP per capita (constant 2015 US\$) and the shares of industry and services in GDP.
- **Demographic and social** aspects are captured by the urban population share and access to electricity, both expressed as percentages of the total population, describing modernization, spatial concentration of demand, and infrastructure development

2.4.3 Excluded Variables

The agricultural share, although initially included, was excluded because the three sectors are complementary by definition and sum to 100% of GDP, meaning that variations in one are mechanically reflected in the others. Retaining only industry and services allows the analysis to focus on the most energy-intensive and structurally dynamic sectors while avoiding redundancy.

Four variables were **excluded** from the final dataset due to limited analytical relevance or conceptual overlap. *CO₂ damage* represents an environmental outcome rather than a driver, *Combustible renewables and waste* duplicates information already captured by total renewable energy and fuel shares, and *Electricity losses* offer little additional insight once energy consumption and efficiency are considered. The exclusion of the agricultural share follows the rationale explained above.

2.4.4 Final Set of Model Variables

This refinement produced a concise and consistent set of covariates representing the main structural domains that are energy demand and efficiency, fuel composition, economic development, and demographic context while ensuring cross-country comparability and avoiding redundancy among indicators.

DESCRIPTIVE ANALYSIS

The descriptive analysis provides an initial overview of the dataset, aimed at identifying the main empirical patterns, assessing data quality, and supporting the subsequent model specification. This stage serves to characterize both the distributional properties of the variables and the relationships among them, highlighting potential issues such as skewness, outliers, or multicollinearity that could affect model performance.

3.1 DISTRIBUTION OF CO₂ EMISSIONS PER CAPITA

Across the full sample (1990–2021), per-capita CO₂ emissions range from nearly 0.04 tons per person in low-income economies to over 30 tons in the highest-emitting countries, with a global mean of around 5.2 tons and a median of approximately 3 tons. This wide dispersion reflects the stark heterogeneity in economic development, energy access, and industrial intensity among countries.

3.1.1 Boxplot Analysis

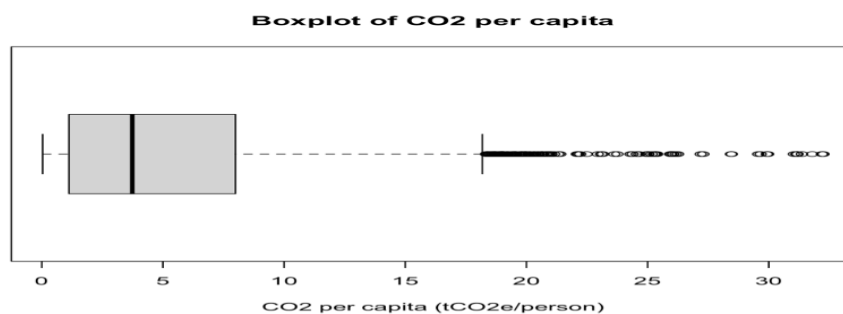


FIGURE 2 - BOXPLOT OF CO₂ PER CAPITA

The boxplot in *Figure 2* illustrates the distribution of per-capita CO₂ emissions across all country-year observations. The median value is around 4-5 tons per person, with most countries falling within an interquartile range between 2 and 7 tons. The distribution is markedly right-skewed, with numerous outliers extending beyond 20 tons per capita, corresponding mainly to industrialized and fossil-fuel dependent economies.

This asymmetry reflects the substantial cross-country heterogeneity in carbon intensity: while most developing countries exhibit low per-capita emissions, a limited number of high-income economies contribute disproportionately to global totals. Such skewness motivates the later adoption of a logarithmic transformation to stabilize variance and improve model interpretability.

3.1.2 Histogram Analysis

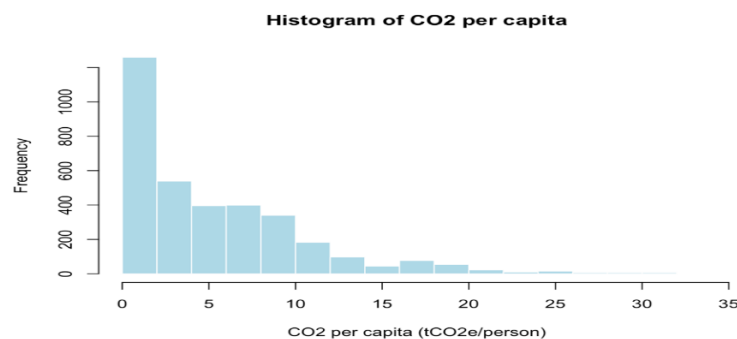


FIGURE 3 - HISTOGRAM OF CO₂ PER CAPITA

The histogram in **Figure 3** confirms the strong right-skewness of per-capita CO₂ emissions already highlighted by the boxplot. The majority of observations cluster below 5 tons per person, while only a small fraction of countries reaches exceptionally high values above 20–30 tons. This distribution underlines the persistent global imbalance in emission intensity between high and low-income economies.

3.2 IMPLICATIONS FOR DATA TRANSFORMATION

Given the pronounced asymmetry and the concentration of values at lower emission levels, a logarithmic transformation of the variable will be applied in the data preparation stage to reduce skewness and improve model performance.

3.3 CORRELATION AND MULTICOLLINEARITY ANALYSIS

3.3.1 Correlation Matrix Overview

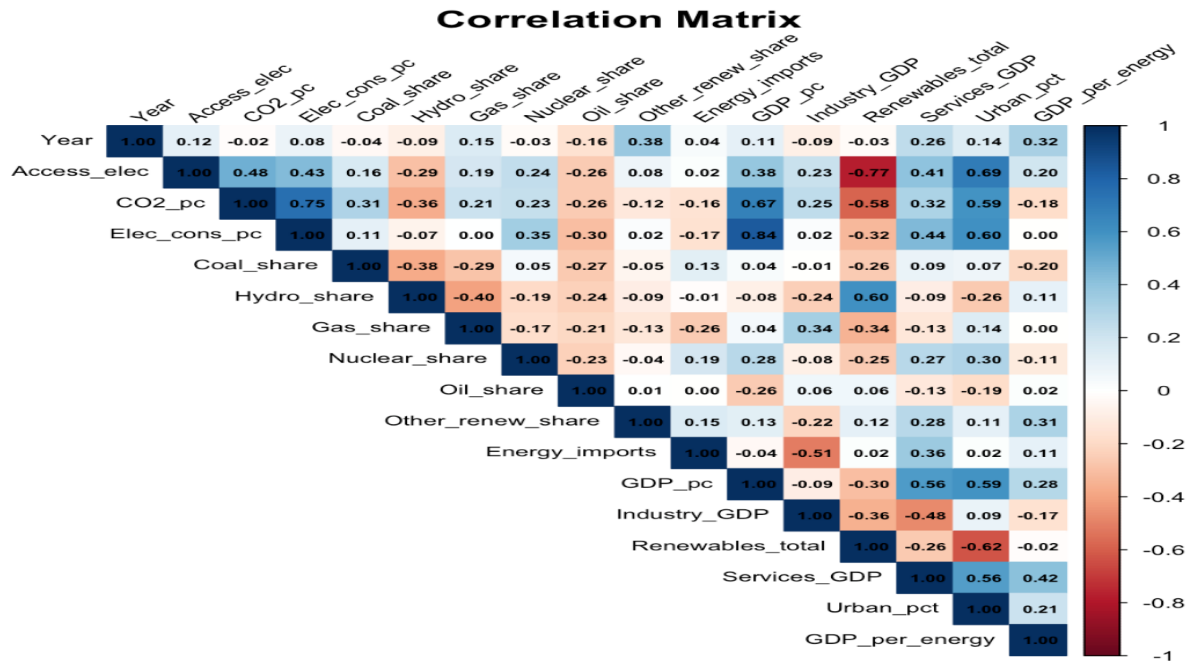


FIGURE 4 - CORRELATION MATRIX

The correlation matrix (**Figure 4**) highlights the main structural relationships among the energy, economic, and demographic variables included in the analysis.

- Income, Energy Use, and Urbanization**
Per-capita CO₂ emissions display the strongest positive associations with electricity consumption per capita ($r \approx 0.75$) and GDP per capita ($r \approx 0.67$), confirming that countries with higher income levels and greater energy demand tend to exhibit higher carbon intensity. Positive correlations also emerge with urban population share ($r \approx 0.59$) and access to electricity ($r \approx 0.48$), suggesting that more urbanized and fully electrified economies are systematically more energy-intensive.
- Fossil Fuels and Nuclear Energy**
From the energy mix perspective, the relationship between CO₂ emissions per capita and fossil fuels is positive for coal ($r \approx 0.31$) and natural gas ($r \approx 0.21$), as expected, while it is negative for oil ($r \approx -0.26$). This latter result reflects the fact that oil-based electricity generation is more common in smaller or less industrialized economies, which overall emit less CO₂ per capita. Nuclear power, on the other hand, shows a moderate positive correlation ($r \approx 0.23$), largely because nuclear generation is concentrated in advanced, energy-intensive countries that already register higher emissions despite the low-carbon nature of the source itself.
- Renewable Energy and Energy Efficiency**

Conversely, renewable energy indicators show negative relationships: both renewable energy consumption ($r \approx -0.58$) and hydropower share ($r \approx -0.36$) are strongly associated with lower emissions per capita, underscoring the mitigating role of renewable energy in national carbon profiles. Energy efficiency, measured as GDP per unit of energy use, also correlates negatively ($r \approx -0.18$), consistent with the idea that economies generating more output per unit of energy tend to have lower carbon intensity.

3.3.2 Synthesis of Correlation Results

Overall, the correlation structure aligns with theoretical expectations: wealthier, urbanized, and fossil-reliant economies account for the highest per-capita emissions, while countries with renewable-oriented or hydro-dominant mixes maintain substantially lower carbon footprints.

DATA PREPARATION FOR EMPIRICAL ANALYSIS

4.1 DATASET PREPARATION STRATEGY

The preparation of the dataset involved a sequence of steps aimed at ensuring that the variables, transformations, and model specifications were adequate for the econometric and machine-learning procedures adopted in this study.

4.1.1 Handling Missing Values

Since standard mixed-effects models (lmer, nlme) do not permit missing values in either fixed or random components, the first step consisted of removing all observations with at least one missing entry among the modelling variables. This complete-case strategy avoids estimation failures or convergence problems and eliminates the need for imputation schemes that would require strong assumptions about within-country temporal dynamics.

4.1.2 Variable Transformations and Standardization

Given the substantial right-skewness observed in the target variable (**CO₂ per capita**) and in two key covariates (**GDP per capita** and **electricity consumption per capita**), each of these indicators was transformed using the natural logarithm of $(x + 1)$. This transformation reduces skewness, stabilizes the residual variance, and improves interpretability in proportional terms. All continuous predictors were subsequently standardized using z-scores to place them on a comparable scale and to enhance numerical stability, especially in the presence of interaction and polynomial terms.

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4.1.3 Construction of Fuel-Presence Dummy Variables

The dataset was further enriched through the creation of **fuel-presence dummy variables**. For each primary energy source (coal, gas, oil, hydro, nuclear, and other renewables), a binary indicator was constructed to capture whether the corresponding share of electricity production exceeded a minimal threshold at least once within the country. These dummies summarize persistent infrastructural or technological features that yearly percentage values alone may not fully capture.

4.1.4 Final Dataset Construction

Once the dataset was cleaned, transformed, and standardized, the variables required for modelling were selected, and the final modelling dataset was constructed. Because mixed-effects models require explicit grouping variables, the country identifier was converted to a factor, and the panel was restricted to observations with complete information across all selected predictors.

4.2 PRELIMINARY LINEAR MODEL DIAGNOSTICS

Before estimating the mixed-effects models, an ordinary least squares regression was fitted exclusively for diagnostic purposes, with the goal of identifying an adequate functional form for the final specification.

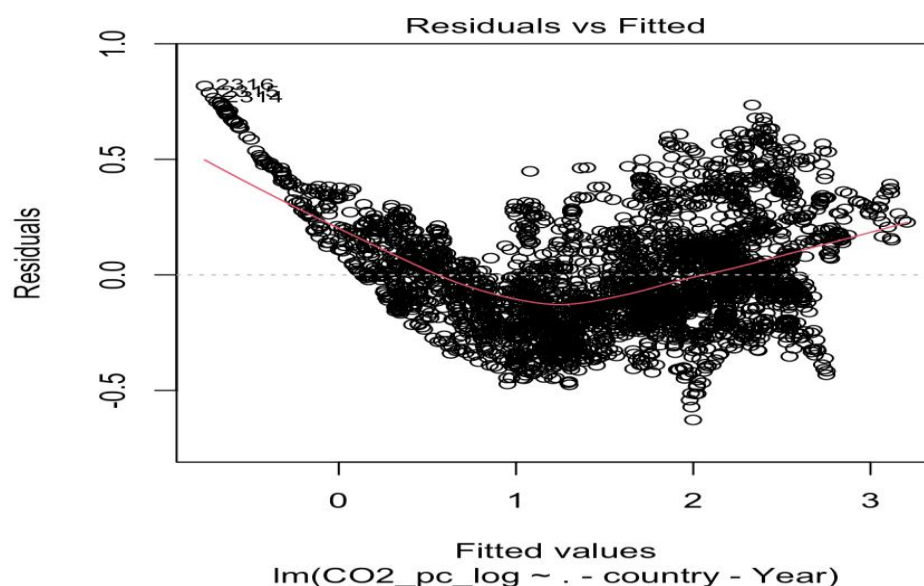


FIGURE 5 - TEST LINEAR MODEL'S RESIDUALS VS FITTED VALUES

The initial linear model revealed clear violations of the linearity assumption: the residual–fitted plot (**Figure 5**) displayed a pronounced U-shaped curvature, indicating that a purely linear structure could not capture the relationship between the log-transformed response and key predictors such as GDP per capita and electricity consumption.

4.2.1 Functional Form Refinement

To address this issue, quadratic terms were introduced for the covariates exhibiting visible curvature. Additional variables were removed because their inclusion generated substantial multicollinearity most notably the log-transformed GDP per capita, which inflated the variance inflation factors to unacceptable levels.

4.2.2 Final Model Specification

The resulting specification, reported below, satisfies all diagnostic requirements (linearity, homoscedasticity, and acceptable multicollinearity levels) and therefore constitutes the functional form adopted for subsequent mixed-effects estimation.

$$\begin{aligned} \log(CO_2_{pc}) = & \log(electricity_consumption_pc) + \log(electricity_consumption_pc)^2 \\ & + GDP_per_unit_of_energy_use + energy_imports + year + urban_pop\% + industry_GDP \\ & + services_GDP + renewable_energy_consumption + coal_share + gas_share + oil_share \\ & + nuclear_share + energy_mix_dummy \end{aligned}$$

This specification produced approximately homoscedastic residuals, no visible curvature, and variance inflation factors within acceptable thresholds, confirming that it satisfies the classical modelling assumptions. The polynomial terms were essential without them; the residuals displayed a clear parabolic pattern indicative of nonlinearity. The exclusion of highly collinear predictors further ensured stable estimation and interpretability in the mixed-effects framework.

4.2.3 Extension to Mixed-Effects Models

The mixed-effects models estimated in the following section directly build upon this validated specification, extending it by incorporating country-specific random effects and serial-correlation structures appropriate for panel data.

METHODOLOGY

After completing the data preparation phase which involved variable selection, transformation, handling missing values, standardization, and feature engineering we applied a structured set of **statistical, econometric, and machine-learning methods** to investigate the determinants of per-capita CO₂ emissions and to classify countries according to their energy and emission profiles.

The methodological framework is explicitly **multi-layered**. Classical regression techniques are first used for interpretability and functional specification, followed by mixed-effects models to address the panel structure of the data. Machine-learning models complement the econometric analysis by capturing non-linearities and interactions, while unsupervised learning techniques are employed to uncover structural groupings of countries. Throughout, strong emphasis is placed on **diagnostics, validation, and robustness**, ensuring the reliability and generalizability of results.

5.1 REGRESSION MODELLING AND FUNCTIONAL SPECIFICATION

5.1.1 Ordinary Least Squares (OLS)

Ordinary Least Squares regression served as an initial benchmark for modelling the relationship between per-capita CO₂ emissions and key explanatory variables such as electricity consumption, GDP per capita, energy efficiency, fuel shares, and demographic indicators.

The baseline model is defined as:

$$y_{it} = \beta_0 + \beta_1 x_{1,it} + \dots + \beta_p x_{p,it} + \varepsilon_{it}$$

with coefficient estimates given by:

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

OLS was not used for final inference but played a **diagnostic role**. Residual–fitted plots and variance inflation factors (VIFs) were inspected to assess linearity, heteroskedasticity, and multicollinearity. These diagnostics motivated subsequent extensions of the model structure.

5.1.2 Polynomial Regression

Exploratory diagnostics revealed that several relationships—most notably those involving electricity consumption and economic scale exhibited clear non-linear patterns. In particular, emissions increased with electricity use at a decreasing rate, suggesting concavity.

To capture this behavior, quadratic terms were introduced:

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \varepsilon$$

Polynomial terms substantially improved residual behaviour, eliminating systematic curvature and yielding a more appropriate functional form for subsequent panel estimation. Only variables displaying clear empirical non-linearity were extended, ensuring model parsimony and interpretability.

5.2 MIXED-EFFECTS MODELS FOR PANEL DATA

5.2.1 Linear Mixed-Effects Model (LME)

Given the longitudinal structure of the dataset repeated observations for each country over time linear mixed-effects models were employed to account for unobserved country-specific heterogeneity.

The general specification is:

$$y_{it} = \beta_0 + \beta_1 X_{it} + u_i + \varepsilon_{it}$$

where:

- u_i is a **country-specific random intercept**, capturing time-invariant national characteristics (institutional, technological, or behavioural),
- β coefficients represent global effects common across countries.

This structure separates **within-country dynamics** from **between-country differences**, preventing biased standard errors and misleading inference typical of pooled regressions.

In addition to accounting for repeated observations, the inclusion of country-specific random intercepts explicitly captures **unobserved heterogeneity** across countries. These random effects represent time-invariant characteristics such as geography, historical infrastructure, long-standing industrial specialization, institutional quality, and cultural energy-use patterns that

systematically influence baseline emission levels but are not directly observable. By modelling these effects explicitly, the mixed-effects framework separates structural cross-country differences from within-country dynamics, preventing biased inference that would arise under pooled regression approaches.

5.2.2 Mixed-Effects Model with AR(1) Temporal Correlation

CO₂ emissions and energy-system variables evolve smoothly over time, leading to serial correlation in regression residuals. To explicitly model this temporal dependence, an AR(1) structure was introduced:

$$\varepsilon_{it} = \rho \varepsilon_{i,t-1} + \eta_{it}$$

Incorporating AR(1) dynamics substantially improved model adequacy, as confirmed by flattened ACF plots, Durbin–Watson statistics close to 2, and well-behaved residual distributions. Beyond correcting residual autocorrelation, the AR(1) structure reflects the **path-dependent nature of CO₂ emissions**. National energy systems evolve gradually due to the long lifespan of infrastructure, capital stock, and technological adoption. As a result, emissions in a given year are strongly influenced by emissions in the preceding period. Modelling this temporal dependence explicitly allows the error structure to mirror the underlying dynamics of emissions processes, improving both statistical validity and substantive realism. This specification constitutes the main inferential model used to answer Research Question 1.

5.3 MACHINE LEARNING MODELS

5.3.1 Random Forest Regression

Random Forest regression was employed to capture complex non-linearities and high-order interactions that linear and mixed-effects models may not fully represent.

The ensemble prediction is defined as:

$$\hat{y} = \frac{1}{B} \sum_{b=1}^B T_b(x)$$

where each T_b is a decision tree trained on a bootstrapped sample.

Random Forests served primarily as:

- a **non-linear predictive benchmark**, and
- an exploratory tool for **feature importance analysis**.

5.3.2 XGBoost (Extreme Gradient Boosting)

XGBoost was applied as a more advanced boosting-based alternative, offering superior predictive performance and fine-grained interpretability.

The objective function is:

$$\text{Obj} = \sum_i l(y_i, \hat{y}_i) + \sum_k \Omega(f_k)$$

where Ω penalizes model complexity to prevent overfitting.

XGBoost consistently outperformed Random Forests in out-of-sample validation. Feature contributions were analysed using **SHAP values**, which enabled direct comparison between machine-learning importance rankings and econometric coefficients.

The strong convergence between the two approaches reinforces the robustness of the identified emission drivers.

The use of SHAP values addresses the common “**black-box**” criticism of machine-learning models by providing transparent, additive explanations of individual predictions. This enables both global interpretation through feature-importance rankings and local interpretation by quantifying how specific variables shift predicted emissions for individual countries. The close alignment between SHAP-based importance and econometric coefficients reinforces the robustness and substantive relevance of the identified emission drivers.

5.4. CLUSTERING AND DIMENSIONALITY REDUCTION

5.4.1 Principal Component Analysis (PCA)

Due to high correlation among energy-system indicators, PCA was applied to reduce dimensionality and construct orthogonal representations of country profiles:

$$Z_k = a_{1k}x_1 + a_{2k}x_2 + \dots + a_{pk}x_p$$

PCA facilitated interpretability and prevented clustering outcomes from being dominated by redundant variables.

5.4.2 K-Means Clustering

K-Means clustering was applied to PCA-transformed country-level averages to identify **structurally distinct energy-profile archetypes**.

The algorithm minimizes within-cluster variance:

$$\min_{c_1, \dots, c_K} \sum_{k=1}^K \sum_{i \in C_k} \|x_i - \mu_k\|^2$$

The number of clusters was selected using the **silhouette criterion**, balancing statistical separation and interpretability. The resulting clusters reveal meaningful differences in demand intensity, fuel composition, efficiency, and emissions outcomes.

5.5. MODEL DIAGNOSTICS AND VALIDATION

5.5.1 Residual Diagnostics

Model adequacy was systematically evaluated using a set of standard residual diagnostics to verify that the underlying assumptions of the regression models were satisfied and to guide model refinement:

- **Residual vs fitted plots**

Residual vs fitted plots were examined to assess linearity and detect potential heteroskedasticity. Initial specifications exhibited visible curvature and non-constant variance, indicating departures from linearity. These patterns motivated the inclusion of polynomial terms and variable transformations to improve functional form and stabilize variance.

- **Q–Q plots for normality**

Quantile–Quantile (Q–Q) plots were used to evaluate the normality of residuals. While minor deviations in the tails were observed which were typical in macro-panel data, the overall alignment with the theoretical quantiles was satisfactory, supporting the validity of inference based on asymptotic approximations.

- **Autocorrelation diagnostics**

Serial correlation was assessed using Autocorrelation Function (ACF) plots and Durbin–Watson statistics. Evidence of temporal dependence in the residuals justified the introduction of an AR(1) error structure in the mixed-effects models, which substantially reduced autocorrelation and improved model fit.

- **Multicollinearity assessment**

Variance Inflation Factors (VIFs) were computed to identify potential multicollinearity among predictors. Variables associated with excessively high VIF values were excluded or reformulated, ensuring numerical stability and reliable coefficient estimation.

- **Diagnostic-driven model refinement**

Insights from these diagnostics directly informed model adjustments, including the adoption of logarithmic transformations, polynomial terms, and AR(1) structures. The final specifications satisfied key assumptions of linearity, homoscedasticity, limited multicollinearity, and temporal independence.

5.5.2 Cross-Validation Framework

To ensure that model performance reflects **true generalisation ability** rather than artefacts of data structure, we implemented a comprehensive cross-validation framework explicitly designed for **panel data** with both temporal and spatial dependence. Standard random train–test splits are inappropriate in this setting, as they may induce information leakage across countries or years, leading to overly optimistic performance estimates. To address these challenges, we adopted three complementary validation strategies that progressively tighten the separation between training and test data.

5.5.2.1 GroupKFold Cross-Validation by Country

The first validation strategy uses **GroupKFold cross-validation**, where the grouping variable is the country identifier. In this setup, all observations belonging to a given country are assigned entirely to either the training set or the test set within a fold. This prevents the model from being trained and tested on different years of the same country simultaneously.

This approach evaluates the model's ability to **generalise across countries**, answering the question of whether the estimated relationships hold for countries that were never observed during training. GroupKFold is particularly relevant in cross-country studies, where structural heterogeneity across nations may be substantial and country-specific patterns could otherwise inflate predictive accuracy if leakage were allowed. Formally, for each fold k , the training set consists of observations from a subset of countries $\mathcal{C}_{\text{train}}^{(k)}$, while the test set contains observations from the complementary subset $\mathcal{C}_{\text{test}}^{(k)}$, with no overlap between the two. This ensures strict spatial independence between training and evaluation samples.

5.5.2.2 Spatio-Temporal Blocked Cross-Validation

While GroupKFold controls for spatial leakage, it does not explicitly prevent temporal leakage if earlier and later periods exhibit strong serial dependence. To address this, we implemented **spatio-temporal blocked cross-validation**, which enforces separation along both the **country** and **time** dimensions.

Under this strategy, models are trained on observations from earlier years and evaluated on subsequent years, ensuring that information from the future is never used to predict the past. This framework closely mirrors real-world forecasting scenarios, where models are applied to future periods based solely on historical data. Specifically, the dataset is partitioned into temporal blocks, and for each validation step, the training set includes all countries observed up to a given cutoff year t_0 , while the test set includes observations from years $t > t_0$. This design simultaneously:

- preserves the panel structure of the data,
- avoids temporal leakage, and
- evaluates model stability under evolving economic and energy-system conditions.

This validation scheme is particularly stringent and provides a realistic assessment of predictive robustness in dynamic settings.

5.5.2.3 Leave-One-Country-Out (LOCO) Validation

As the most conservative evaluation strategy, we implemented **Leave-One-Country-Out (LOCO)** cross-validation. In this framework, the model is trained on the full dataset excluding one country and then tested exclusively on the omitted country. This process is repeated iteratively for each country in the sample.

LOCO validation directly assesses whether the model captures **general structural relationships** rather than country-specific idiosyncrasies. High performance under LOCO indicates that the model is not merely interpolating within known national trajectories but is instead learning patterns that transfer meaningfully across institutional, economic, and energy contexts.

Because each test fold consists of a single country, this approach represents a severe stress test for model generalization and provides strong evidence regarding the external validity of the results.

5.5.3 Evaluation Metrics

Across all validation strategies, model performance was evaluated using three complementary metrics:

- **Root Mean Squared Error (RMSE)**, which penalizes large prediction errors and is sensitive to outliers;
- **Mean Absolute Error (MAE)**, which provides a robust measure of average deviation between predicted and observed values;
- **Coefficient of Determination (R^2)**, which measures the proportion of variance in per-capita CO₂ emissions explained by the model.

Using multiple metrics ensures that performance assessments are not driven by a single criterion and allows for a nuanced comparison across modelling approaches.

5.5.4 Interpretation and Rationale

The combination of GroupKFold, spatio-temporal blocked cross-validation, and LOCO provides a **hierarchy of increasingly demanding validation schemes**, each targeting a different source of potential bias. Consistent performance across these

strategies indicates that predictive accuracy is not driven by information leakage across countries or years, but instead reflects stable and transferable relationships.

This multi-layered validation framework strengthens confidence in both the econometric and machine-learning results and supports the claim that the identified drivers of per-capita CO₂ emissions generalise reliably across space and time.

RESULTS

6.1 RQ1: WHAT ARE THE KEY DRIVERS OF PER-CAPITA CO₂ EMISSIONS?

6.1.1 Model Specification and Adequacy

The mixed-effects model with AR (1) autocorrelation provides a comprehensive explanation of the structural determinants of per-capita CO₂ emissions across 104 countries from 1990 to 2021. Before interpreting the coefficients, it is necessary to assess model adequacy.

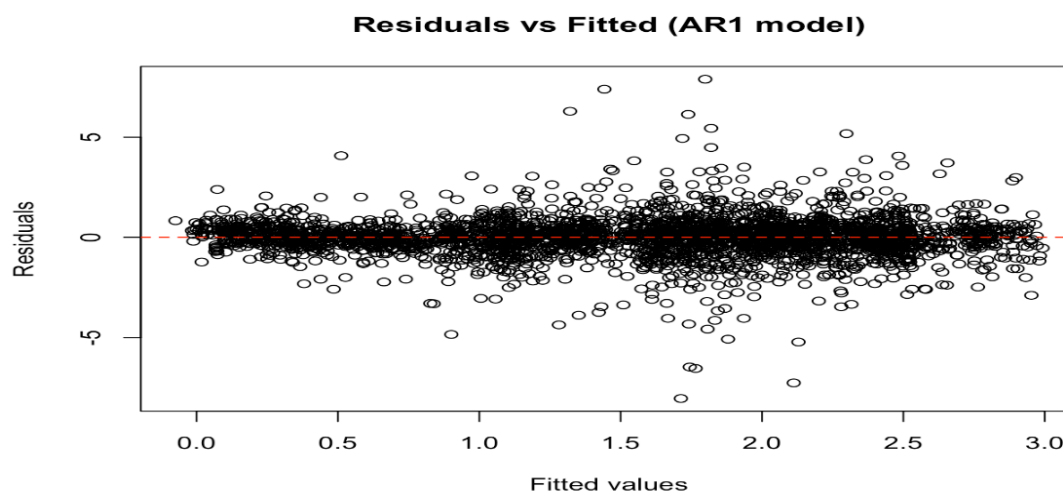


FIGURE 6 - RESIDUAL VS FITTED VALUES AR(1) MODEL

The residual–fitted plot (**Figure 6**) shows a homogeneous dispersion of residuals around zero with no visible curvature or funnel shape, indicating that the logarithmic transformation of the dependent variable and the quadratic specification of electricity consumption successfully removed the non-linear patterns observed in the preliminary OLS model.

6.1.2 Interpretation of the AR(1) Structure

Beyond correcting residual autocorrelation, the AR(1) specification reflects the **inherent inertia of national energy systems**. Electricity generation, transport networks, and industrial capital have lifespans of several decades, implying that current emissions are tightly constrained by past infrastructure investments. As a result, national CO₂ emissions are **path-dependent** and cannot adjust instantaneously to policy or economic changes. The AR(1) structure therefore captures not only statistical dependence but also the socio-economic reality of emissions “lock-in,” ensuring that the model reflects the slow dynamics of real-world energy transitions.

6.1.3 Residual Diagnostics and Error Structure

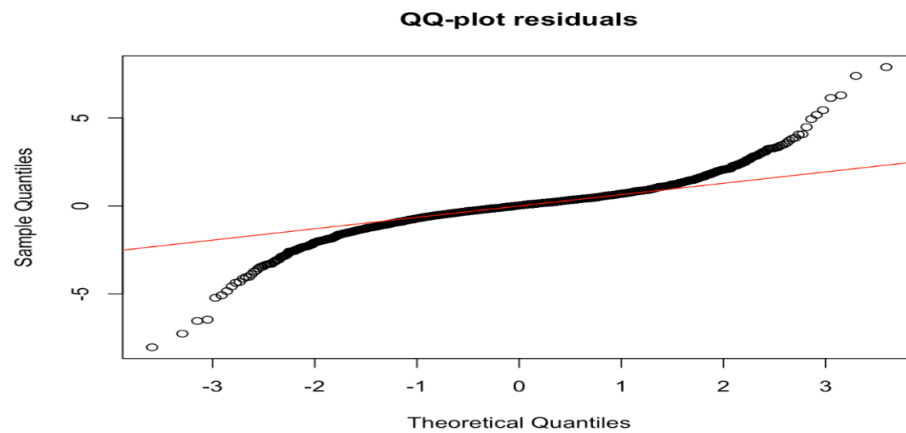


FIGURE 7 - QQ PLOT AR(1) MODEL

Residuals also exhibit approximate normality, as illustrated by the QQ-plot (Figure 7), with moderate deviations in the tails that are typical for macro-panel data but do not compromise inference.

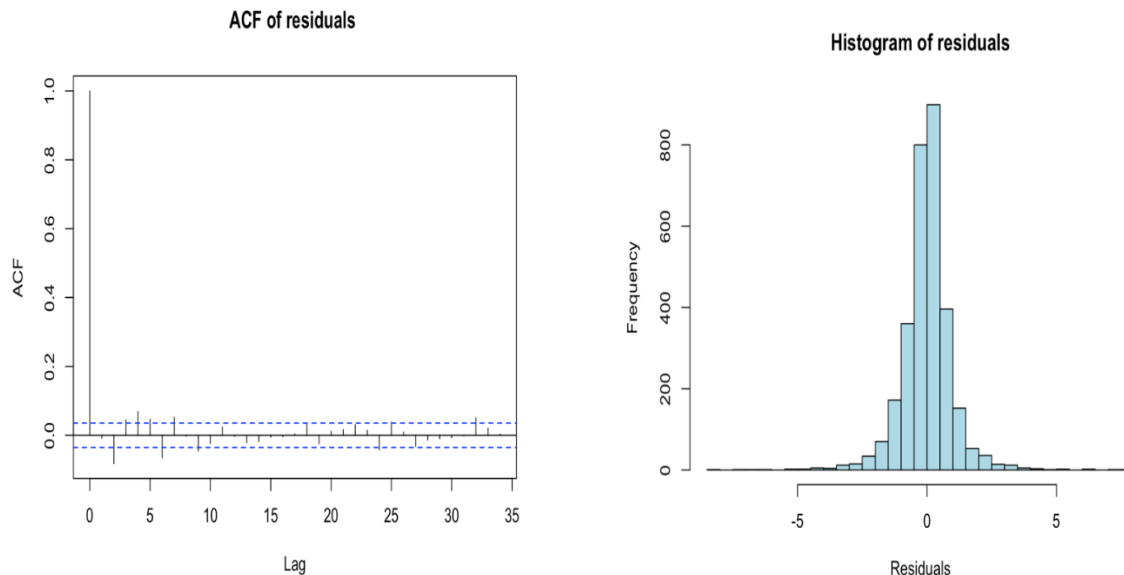


FIGURE 8 - AUTOCORRELATION DIAGNOSTIC

FIGURE 9 - RESIDUALS DISTRIBUTION AR(1) MODEL

Serial correlation is fully eliminated by the AR (1) structure: the ACF of residuals (Figure 8) is essentially flat, and the Durbin–Watson statistic ($DW \approx 2.01$) confirms the absence of systematic temporal dependence. The histogram of residuals (Figure 9) further shows a symmetric, centered distribution, consistent with a well-specified error structure.

6.1.4 Effects of Electricity Consumption

With the model validated, the estimated coefficients reveal a coherent and theoretically consistent structure of drivers. Since the dependent variable is in logarithms and the predictors are standardized, coefficients measure semi-elasticities per standard-deviation change. Electricity consumption emerges as the dominant factor: a one-standard-deviation increase in log electricity use raises log CO₂ emissions by approximately 0.56 ($p < 0.001$). Substantively, this implies that moving from a median to an upper-quartile level of electricity demand can increase per-capita emissions by roughly 70–75%, even before accounting for the concavity captured by the positive quadratic term (+0.16). This diminishing marginal effect is consistent with the behavior of highly electrified, structurally efficient economies.

6.1.5 Effects of the Energy Mix

The composition of the energy mix also displays strong predictive capacity. Coal is the most emission-intensive source: a one-standard-deviation increase in its share corresponds to a 7.4% rise in per-capita emissions. Oil and natural gas have more

moderate but still positive effects (+2.6% and +1.4%, respectively). These magnitudes quantify the persistent carbon intensity of fossil-dependent power systems. In contrast, renewable energy deployment exerts the strongest mitigating force in the system: a one-standard-deviation increase in renewable energy consumption reduces per-capita emissions by approximately 28.5% ($p < 0.001$). Energy efficiency, measured through GDP per unit of energy use, further contributes a sizable reduction (−8.8%). Nuclear generation, though less prevalent globally, provides an additional modest but statistically clear decrease of around 4.3%.

6.1.6 Economic Structure and Demographic Effects

Economic structure and demographic context also play meaningful roles. A larger industrial base is associated with slightly higher emissions (+0.85%), while the expansion of the services sector exerts a small negative effect (−1.0%). Urbanization increases emissions (coefficient +0.086), reflecting the concentration of economic and energy-intensive activities within metropolitan environments. The time trend is negative (−0.025 per SD), indicating that countries have experienced a gradual structural decline in emissions once energy and economic characteristics are held constant.

6.1.7 Feature Importance

Feature importance summarises the relative influence of each predictor, helping identify the key structural factors shaping national CO₂ emission levels.

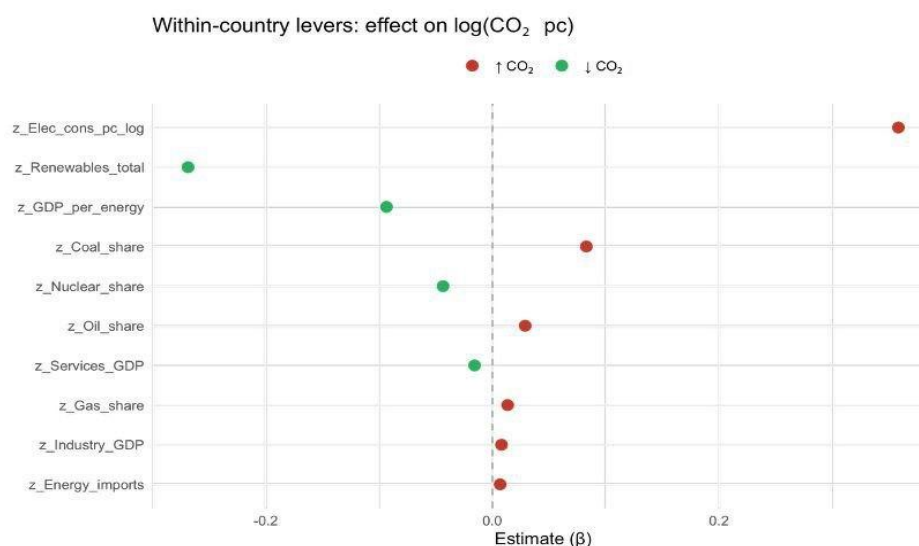


FIGURE-10 - FEATURE IMPORTANCE AR(1) MODEL

Figure-10 presents the standardized coefficients from the AR(1) mixed-effects model, showing the relative contribution of each predictor to per-capita CO₂ emissions after accounting for country-specific effects and temporal autocorrelation. Positive values indicate that higher levels of the variable are associated with higher emissions, while negative values correspond to mitigating effects.

6.1.8 Model Performance and Predictive Validation

The model explains an exceptionally high share of cross-country and temporal variation. Both the marginal and conditional R^2 equal 0.93, indicating that once autocorrelation is corrected, fixed effects alone account for nearly all observed variability. Predictive validation confirms this robustness: grouped 5-fold cross-validation yields an R^2 of 0.93 and RMSE of 0.21; spatio-temporal blocked validation produces $R^2 = 0.94$ and RMSE = 0.18; and leave-one-country-out validation yields similar accuracy ($R^2 \approx 0.94$, RMSE ≈ 0.20). These results confirm that the model generalizes well across both space and time.

6.1.9 Summary of Results

Overall, the findings portray a consistent energy emissions system has high electricity demand, urbanisation, and dependence on fossil fuels systematically increase per-capita CO₂ emissions, while renewable energy, structural energy efficiency, and nuclear power have large mitigating effects. The estimated magnitudes are substantial, interpretable, and aligned with theoretical expectations, confirming that the model successfully captures the fundamental drivers shaping national carbon footprints.

6.2 RQ2: PREDICTIVE CAPACITY OF MACHINE-LEARNING MODELS

6.2.1 Motivation for Machine-Learning Analysis

To complement the inferential mixed-effects analysis, we evaluate how well non-linear machine-learning models like Random Forests and XGBoost can predict per-capita CO₂ emissions. The rationale for this second modelling strategy is twofold. First, machine-learning models can detect complex interaction patterns and non-linearities that linear econometric specifications, even augmented with quadratic terms, cannot fully capture. Second, predictive evaluation provides an external check of the structural model: if the variables identified as key determinants in the mixed-effects model are also the dominant predictors in ML models, this convergence substantially strengthens the credibility of our empirical findings.

6.2.2 Model Setup and Validation Strategy

Both algorithms were estimated using the same set of predictors as in the AR(1) mixed-effects model. Out-of-sample performance was assessed using three rigorous validation schemes designed to prevent spatial and temporal leakage which are GroupKFold by country, spatio-temporal blocked cross-validation, and leave-one-country-out (LOCO) validation. Across all schemes, XGBoost consistently achieved the highest predictive accuracy, with out-of-sample coefficients of determination between 0.91 and 0.94 and RMSE in the range 0.18–0.21, outperforming Random Forests by a small but systematic margin

6.2.3 Predictive Performance Diagnostics

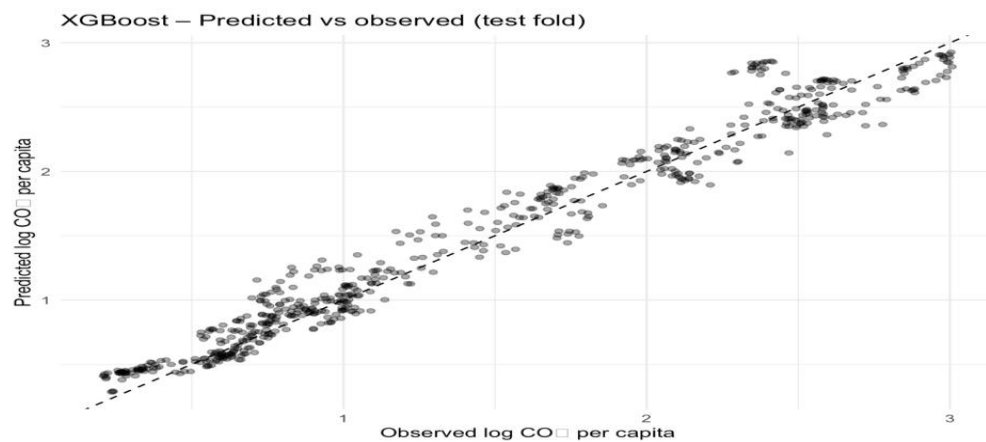


Figure 11- ML-A: Scatter plot Predicted vs Observed

The predicted–observed scatter plot (**Figure 11 ML-A**) reveals an almost perfect alignment with the 45-degree line, indicating that the model produces unbiased predictions across the full spectrum of emission levels.

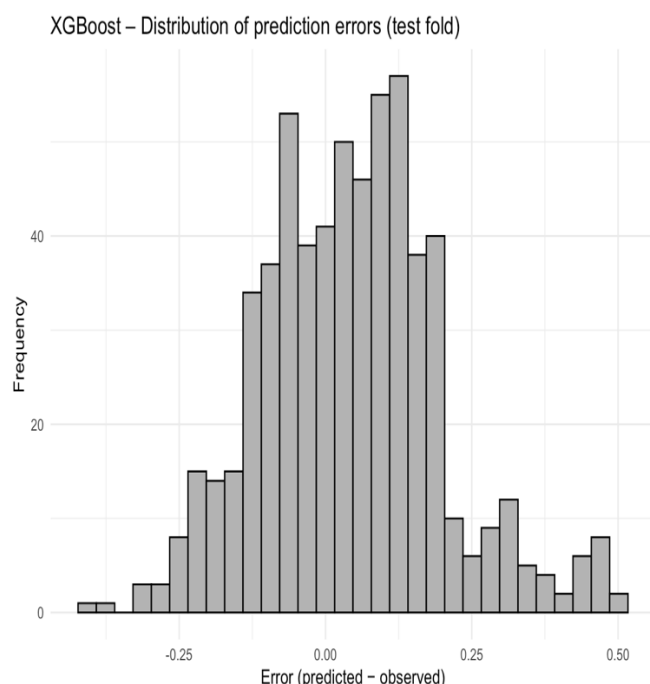


Figure 12 - ML-B: Error histogram

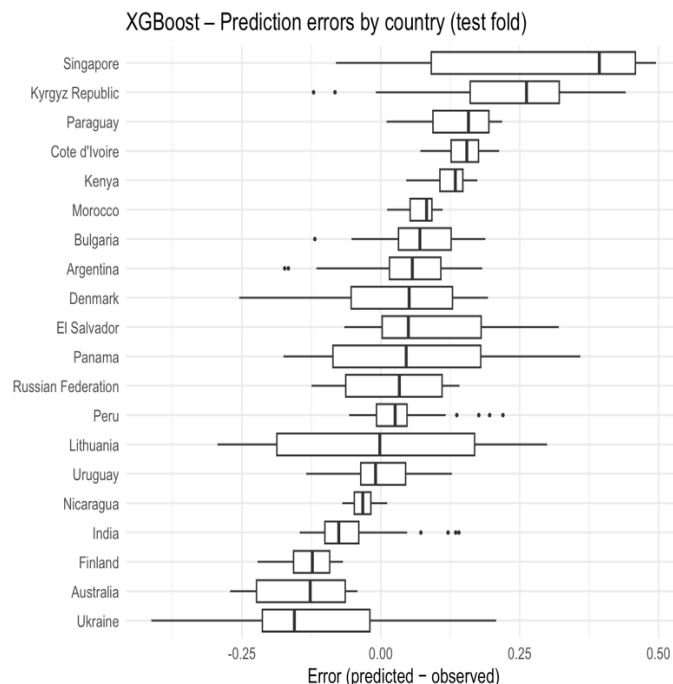


Figure 13 - ML-C: Prediction errors by country

The distribution of residuals (**Figure 12 - ML-B**) is symmetric and centered around zero, suggesting the absence of systematic over- or under-prediction. A country-level decomposition (**Figure 13 - ML-C**) shows that mispredictions are concentrated in a small subset of structurally volatile economies such as Singapore, Paraguay, or El Salvador while predictive accuracy remains high and stable for the vast majority of countries, including both high-income and emerging economies.

6.2.4 Feature Importance and SHAP Analysis

Beyond prediction accuracy, a central advantage of XGBoost is its ability to provide rich interpretability through SHAP values, which quantify how each feature contributes to individual predictions.

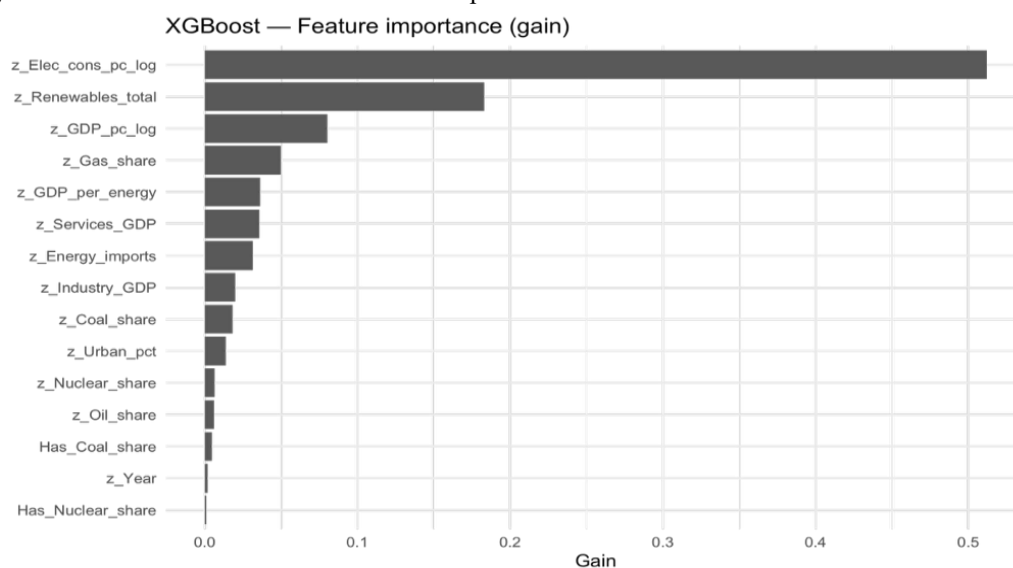


Figure- 14 ML-D: SHAP plot showing feature importance

The mean absolute SHAP plot (**Figure-14 ML-D**) demonstrates a striking consistency with the mixed-effects results: log electricity consumption per capita is by far the most influential predictor, followed by renewable energy consumption, GDP

per unit of energy use, and the share of coal in electricity generation. This hierarchy mirrors the structure uncovered in RQ1, confirming that the primary determinants of emissions are both statistically and predictively fundamental.

6.2.5 Non-Linear Effects and Threshold Behaviour

To examine non-linear patterns, we analyse SHAP profiles for the main drivers and find clear threshold-like changes in the strength (and in one case the direction) of their contributions.

- **Electricity Demand Threshold Effects**

First, electricity demand displays a marked amplification at higher levels: for $z_Elec_cons_pc_log$, the estimated breakpoint is close to the mean of the standardized distribution (≈ 0.006), and the fitted slope increases sharply from ≈ 0.55 below the breakpoint to ≈ 6.14 above it.

This indicates that additional electricity consumption becomes disproportionately associated with higher predicted CO₂ once countries are in a higher-demand regime.

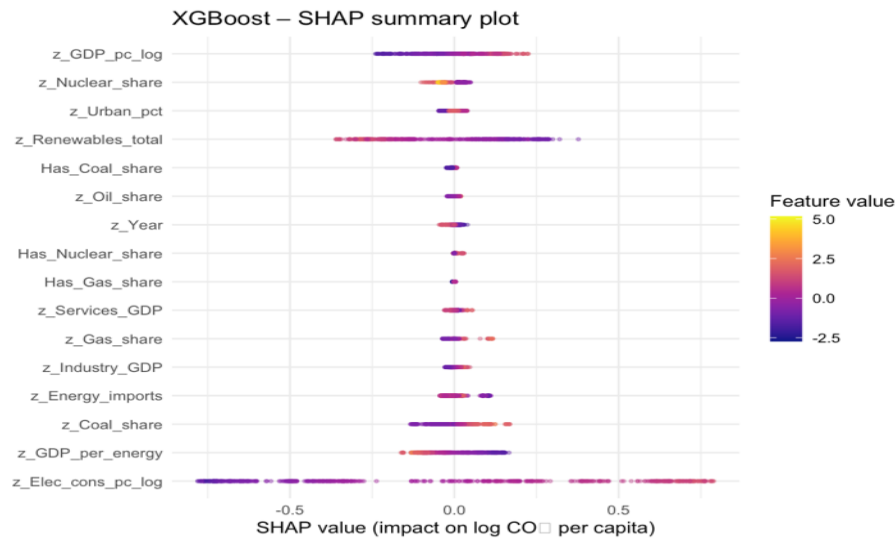


Figure-15 ML-E: SHAP summary plot

- **Energy Productivity and Diminishing Returns**

Second, energy productivity shows the opposite pattern, consistent with diminishing returns. For $z_GDP_per_energy$, SHAP values decline steeply up to a breakpoint around ≈ 0.427 (slope ≈ -0.91) and then flatten substantially (slope ≈ -0.09), implying that efficiency improvements are associated with large emission reductions at low-to-medium productivity levels but yield smaller marginal gains among already efficient observations (**Figure-15 ML-E**).

- **Urbanisation Regime Shifts**

Third, urbanisation exhibits a regime shift in sign: for z_Urban_pct , the breakpoint is ≈ 0.706 and the slope switches from mildly positive to negative ($\approx +0.16 \rightarrow -0.33$).

This suggests that the association between urbanisation and emissions is not uniform across the distribution, and that highly urbanised observations are characterised by a different marginal pattern than low-urbanisation ones.

6.2.6 Cross-Variable Interpretation of Non-Linear Regimes

Finally, we relate these regimes to cross-country profiles by classifying countries according to how persistently they lie above each breakpoint over time. For electricity demand, countries such as Denmark, Finland, France, Italy, and Korea are consistently above the threshold, whereas countries such as Ghana, Guatemala, Honduras, and India remain consistently below, reflecting persistent differences in electricity demand intensity.

For urbanisation, highly urbanised economies such as Singapore and the UAE are consistently above the threshold, while a much larger set of countries stays below for most years.

These descriptive splits help anchor the non-linear patterns in interpretable cross-country regimes, without changing the main conclusion: electricity demand, energy-system composition, and energy productivity remain the core drivers shaping per-capita CO₂ emissions.

6.2.7 Summary of Results

Overall, the machine-learning analysis confirms and strengthens the conclusions of RQ1. XGBoost achieves high predictive accuracy, identifies the same hierarchy of drivers, reveals interpretable non-linear patterns consistent with theoretical expectations, and generalizes robustly across both time and countries. ML models therefore serve not only as predictive tools but also as a powerful validation instrument for the mixed-effects framework, providing convergent evidence on the central role of electricity demand, fossil-fuel dependency, energy efficiency, and renewable deployment in shaping national CO₂ emission trajectories.

6.3 RQ3: COUNTRY DEVIATIONS (“FAIR RESIDUALS”)

6.3.1 Definition and Interpretation of Fair Residuals

To investigate cross-country heterogeneity beyond observable structural drivers, we analyse country-level deviations from the predictions of the fixed-effects component of the AR(1) mixed-effects model. These deviations, hereafter referred to as *fair residuals*, are defined as the difference between observed per-capita CO₂ emissions and those predicted solely by the model’s structural covariates, excluding country-specific random intercepts. By construction, fair residuals capture whether a country emits systematically more or less CO₂ than expected, conditional on its energy mix, economic structure, urbanization, and level of development.

6.3.2 Global Distribution of Fair Residuals

The global distribution of country-year fair residuals is shown in **Figure RQ3-A**. The distribution is centered close to zero and approximately unimodal, indicating that the fixed-effects specification provides a well-calibrated baseline on average at global level. Nevertheless, the presence of a pronounced right tail reveals a subset of observations in which emissions substantially exceed structural expectations, motivating a closer country-level examination. The overall dispersion of fair residuals is summarized in **Figure RQ3-B**, which shows moderate variability with limited extreme outliers, suggesting that cross-country differences are suggesting structured heterogeneity rather than a handful of extreme anomalies.

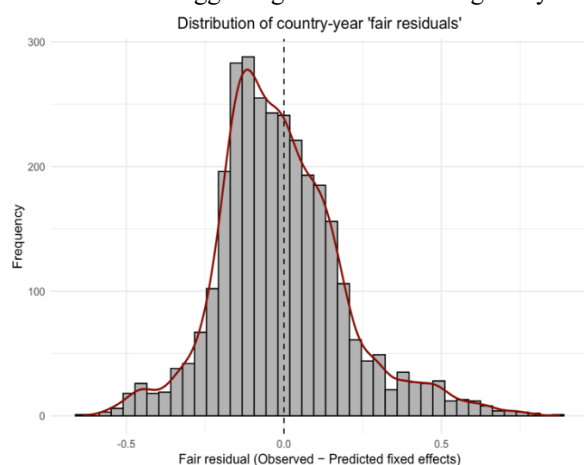


Figure RQ3-A

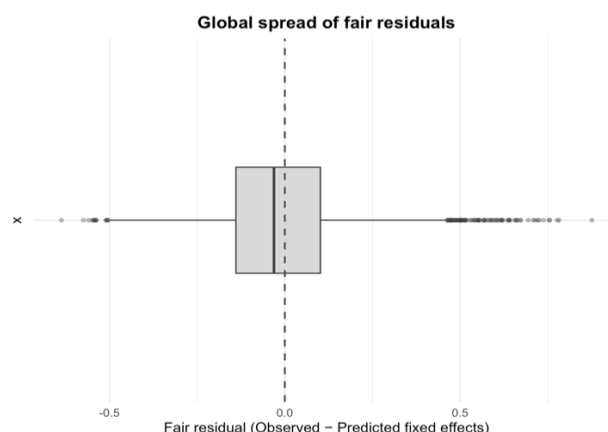


Figure RQ3-B

6.3.3 Average Country-Level Deviations

Aggregating fair residuals at the country level highlights persistent patterns of over- and under-performance. **Figure RQ3-C** reports mean country-level deviations with 95% confidence intervals. Several countries exhibit statistically significant positive residuals, indicating emissions consistently above model-predicted levels even after accounting for structural drivers. These include Oman, Turkmenistan, Nigeria, Estonia, and the United Arab Emirates, as well as high-income economies such as Luxembourg and Ireland. Conversely, countries such as Sweden, the Kyrgyz Republic, Morocco, Jordan, and the Philippines display significantly negative residuals, emitting systematically less CO₂ per capita than predicted by their observable

characteristics. Several country means are statistically distinguishable from zero, indicating systematic average deviations from the baseline but reflect persistent country-level deviations.

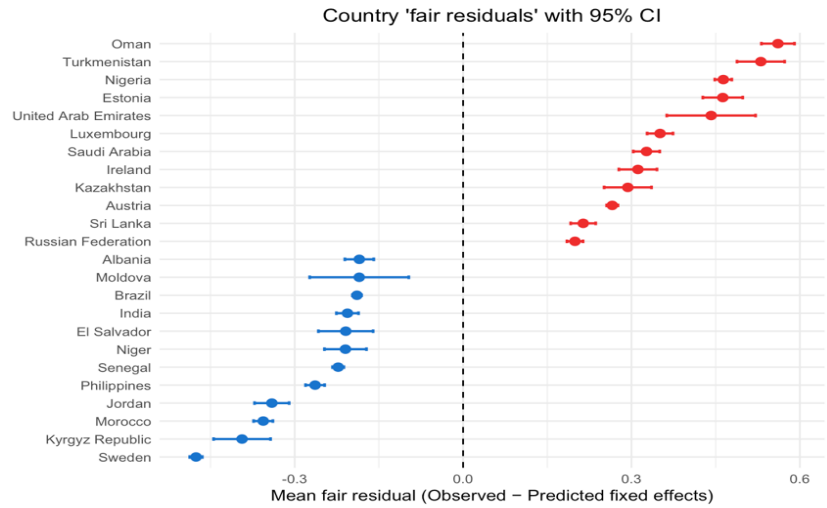


Figure RQ3-C

6.3.4 Temporal Persistence of Country Deviations

To assess the temporal stability of these patterns, **Figure RQ3-D** presents average fair residuals by country and decade. The heatmap reveals strong persistence over time: countries identified as structurally above- or below-expected tend to maintain their relative positions across decades, although the magnitude of deviations varies. This temporal consistency suggests that fair residuals capture enduring institutional, technological, or behavioral factors not explicitly modelled in the fixed-effects specification.

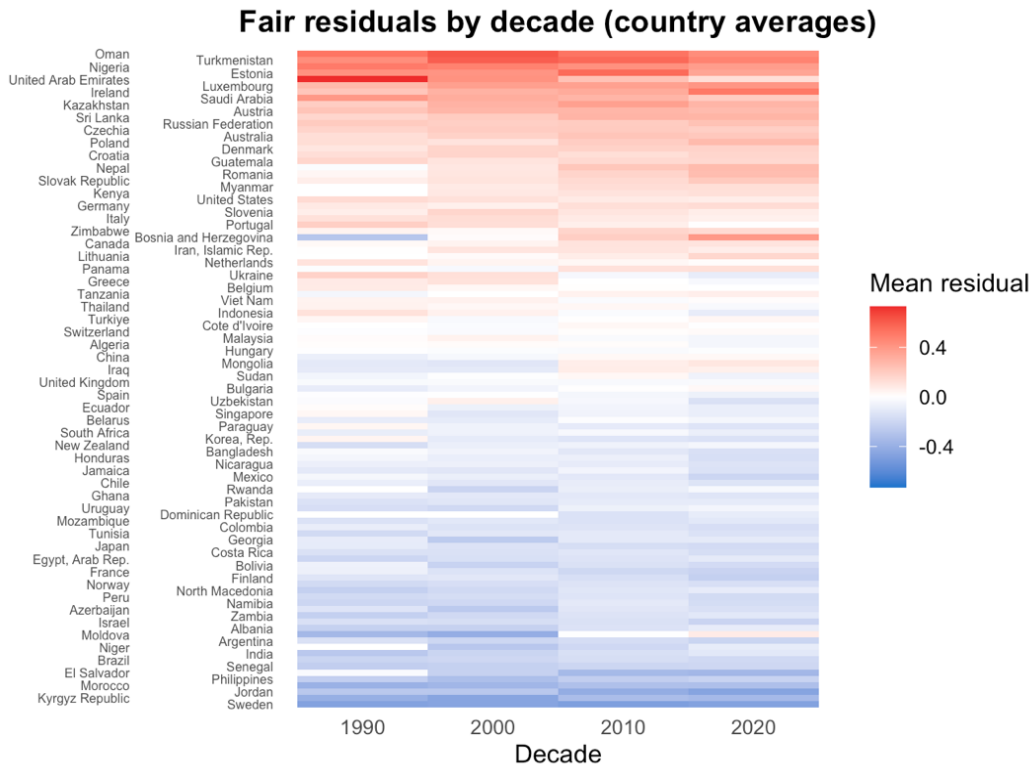


Figure RQ3-D

6.3.5 Mean–Volatility Relationship of Fair Residuals

Beyond average performance, countries also differ in the temporal stability of their deviations from structural expectations. **Figure RQ3-E** plots each country's mean fair residual against the within-country standard deviation of those residuals. The horizontal axis captures the average degree of over- or under-emission relative to predicted levels, while the vertical axis measures the volatility of this deviation over time.

6.3.6 Interpretation of Country-Level Patterns

Interpreting fair residuals at the country level reveals persistent patterns of over- and under-performance. For instance, **Sweden** exhibits consistently negative residuals, emitting less CO₂ than predicted due to its hydro- and nuclear-based energy mix and long-standing efficiency policies. In contrast, countries such as the **United Arab Emirates, Oman, and Turkmenistan** display persistent positive residuals, indicating emissions that exceed structural expectations. These deviations suggest that income and electricity demand alone do not fully explain national carbon footprints and point to deeper institutional and policy-related factors.

6.3.7 Stability versus Structural Inefficiency

The dispersion of points reveals no systematic relationship between mean deviation and volatility. Countries that consistently emit more or less than expected often display relatively stable residuals, indicating persistent but predictable departures from structural benchmarks. Conversely, several countries with mean deviations close to zero exhibit high volatility, alternating between periods of over- and under-performance. This pattern indicates that structural inefficiency and temporal instability are distinct dimensions of emissions behavior: persistent over- or under-emission does not imply instability, and stability does not imply structural efficiency.

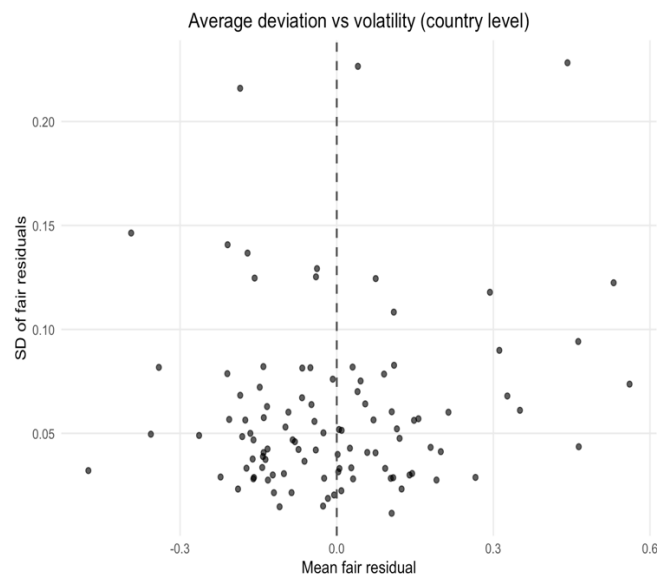


Figure RQ3-E

6.3.4 Summary of Results

Overall, the results of RQ3 reveal substantial and persistent cross-country heterogeneity in per-capita CO₂ emissions that cannot be explained by observable structural drivers alone. Some countries systematically outperform model expectations, while others emit significantly more than predicted, even after controlling for energy use, fuel composition, economic structure, and temporal dynamics. These deviations point to the importance of unobserved factors such as policy frameworks, technological adoption, institutional quality, or behavioral patterns.

6.4 RQ4: CLUSTERING OF ENERGY PROFILES

6.4.1 Motivation and Clustering Strategy

To investigate whether countries can be grouped into distinct energy-profile archetypes, we performed a clustering analysis on country-level average profiles computed over the most recent period, 2015–2021. This time window was deliberately chosen to characterize contemporary energy-system configurations: prior to 2015, many countries were still undergoing more pronounced transitional dynamics that could blur stable archetypes and reduce cluster interpretability. Each country is represented by a vector of structural and energy-system indicators capturing electricity demand intensity (electricity consumption per capita), supply composition (shares of coal, gas, renewables, and nuclear in electricity generation), and macro-efficiency and economic structure (GDP per unit of energy use and services value added). All variables were standardized prior to clustering. To ensure comparability and mitigate biases from incomplete data coverage, country profiles were computed as averages over country–year observations with complete information across all clustering variables, and we retained only countries with a minimum number of complete years within the 2015–2021 window.

6.4.2 Cluster Selection and Validation

The number of clusters was selected using the average silhouette criterion and, as shown in **Figure RQ4-A**, the silhouette index peaks at $k = 6$ (≈ 0.30), indicating moderate but meaningful separation and supporting a six-cluster solution that balances statistical fit and interpretability.

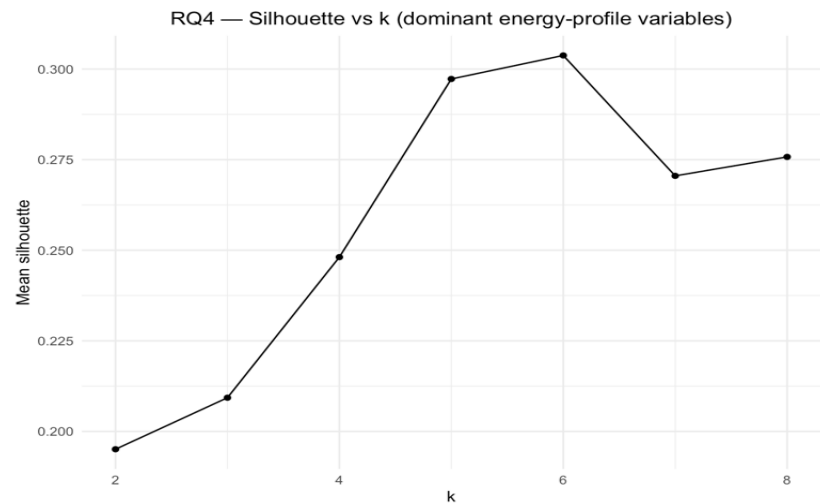


Figure RQ4-A

6.4.3 Energy-Profile Archetypes

The resulting partition reveals six distinct energy-profile archetypes.

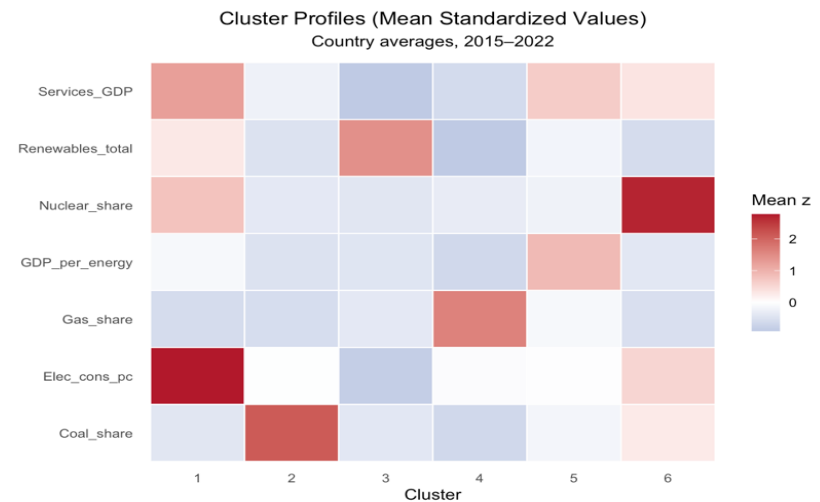


Figure RQ4-B

Their characteristic feature signatures are summarized in **Figure RQ4-B (heatmap)**, which reports mean standardized values by cluster and highlights the dominant dimensions distinguishing groups.

- **High-Demand, Low-Carbon Supply Systems (Cluster 1)**

Cluster 1 is characterized by very high electricity consumption per capita, relatively low coal reliance, and a comparatively cleaner supply mix. This group includes highly electrified and advanced economies such as France, Sweden, Finland, Norway, Switzerland, and Canada, where high demand coexists with substantial nuclear, renewable, or low-carbon generation.

- **Coal-Dependent Energy Systems (Cluster 2)**

Cluster 2 corresponds to coal-dependent energy systems, with coal representing the dominant source of electricity generation. Representative countries in this group include China, Poland, South Africa, Bosnia and Herzegovina, and India, reflecting persistent reliance on coal despite heterogeneous income levels.

- **Low-Demand, Renewable-Intensive Systems (Cluster 3)**

Cluster 3 combines low electricity demand with very high renewable shares, capturing primarily lower-income or emerging economies where renewables account for a large fraction of energy supply. Examples include Nepal, Rwanda, Zambia, Mozambique, Guatemala, and Zimbabwe.

- **Gas-Dominated Energy Systems (Cluster 4)**

Cluster 4 is defined by gas-dominated electricity systems, with exceptionally high gas shares and very limited renewable penetration. This archetype includes fossil-rich or gas-exporting countries such as Oman, United Arab Emirates, Saudi Arabia, Algeria, and Turkmenistan.

- **Diversified and Efficiency-Oriented Systems (Cluster 5)**

Cluster 5 represents a more diversified and efficiency-oriented profile, characterized by moderate electricity demand, mixed supply composition, and the highest levels of energy efficiency (high GDP per unit of energy use). Typical members include Germany, United Kingdom, Netherlands, Spain, Italy, Japan, and Portugal, often associated with service-oriented economies and diversified energy mixes.

- **Fossil-Tilted, Intermediate-Demand Systems (Cluster 6)**

Finally, Cluster 6 captures countries **Figure RQ4 – C** with above-average electricity demand combined with a more fossil-tilted structure, though less extreme than coal-dependent systems. Examples include Czechia, Bulgaria, Slovenia, Hungary, Estonia, and Korea, reflecting intermediate positions between coal-heavy and diversified profiles.

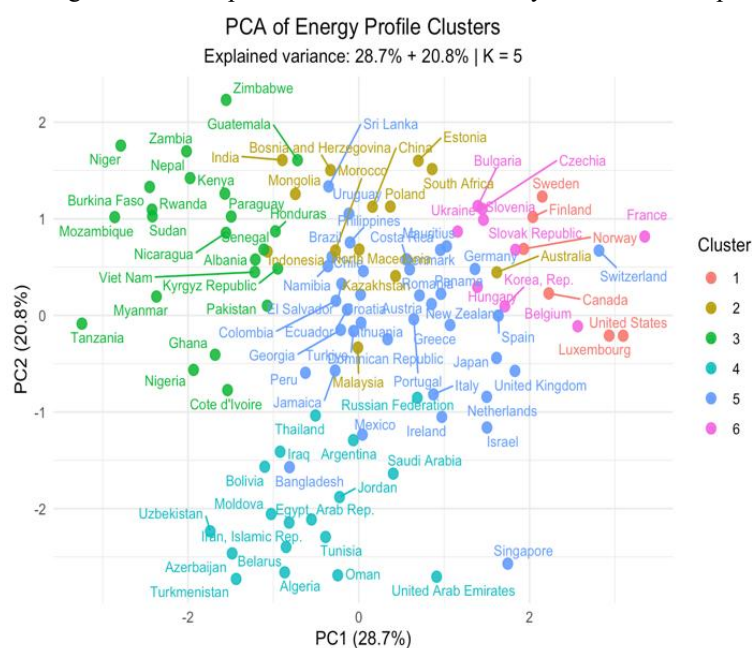


Figure RQ4-C

6.4.4 PCA Representation of Cluster Structure

For visual intuition, **Figure RQ4-D** projects countries onto the first two principal components of the standardised feature space ($\approx 49.5\%$ cumulative variance). While some overlap between clusters is expected due to dimensionality reduction and the continuous nature of energy transitions, the PCA representation confirms meaningful structure aligned with electricity demand intensity and fuel composition gradients.

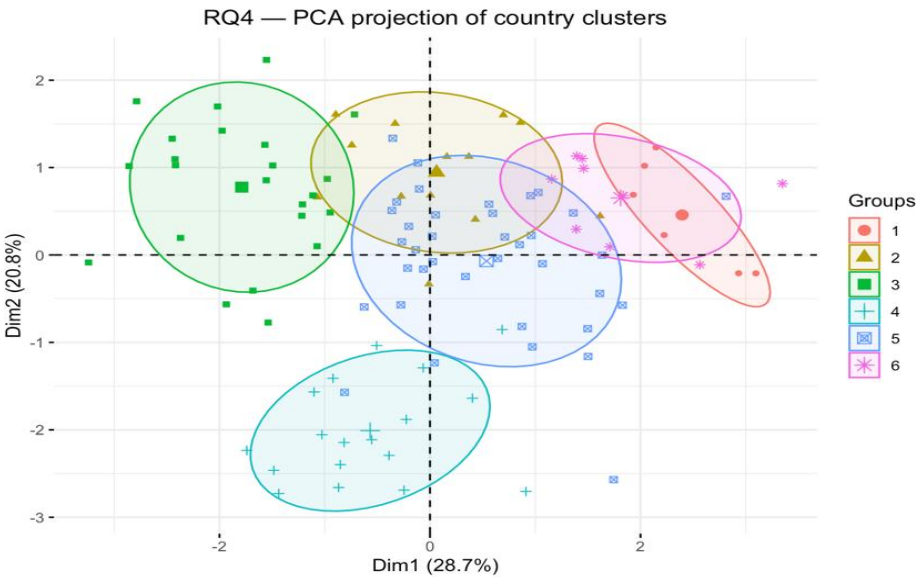


Figure RQ4-D

6.4.5 Emissions implications of the energy archetypes.

To connect the identified energy-profile archetypes to their climate outcomes, we compare per-capita CO₂ emissions across clusters over the same contemporary window used for clustering (2015–2021).

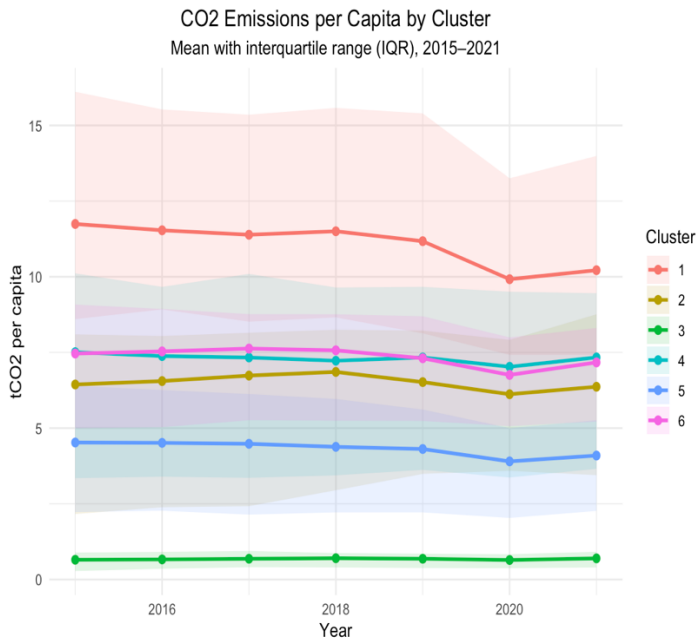


Figure RQ4-E

cluster	mean_co2
1	11.1
6	7.35
4	7.31
2	6.51
5	4.32
3	0.674

Table RQ4-1

Figure RQ4-E reports cluster-level emissions trajectories, while **Table RQ4-1** summarizes the distribution of emissions by cluster. A clear stratification emerges: Cluster 1 exhibits by far the highest per-capita emissions (mean ≈ 11.1 tCO₂), whereas

Cluster 3 remains at very low levels (mean ≈ 0.67 tCO₂). Inter mediate regimes are observed for clusters 6 and 4 (≈ 7.3 tCO₂) and for Cluster 2 (≈ 6.5 tCO₂), and for the more diversified and efficiency-oriented Cluster 5 (≈ 4.3 tCO₂).

6.4.6 Structural Interpretation of Emissions Differences

This ordering is consistent with the structural profiles that define each archetype (**Figure RQ4-D**). Cluster 1 combines extremely high electricity consumption per capita ($\approx 15,647$ kWh) with a comparatively cleaner generation mix, indicating that emissions are primarily driven by a scale effect, very high demand intensity and high-income consumption patterns, rather than by coal dependence. By contrast, Cluster 6 displays above-average electricity demand ($\approx 6,346$ kWh) alongside a more fossil-tilted composition, leading to persistently high emissions despite lower demand than Cluster 1. At the opposite end, Cluster 3 pairs very low electricity demand (≈ 578 kWh) with very high renewable shares ($\approx 63\%$), yielding the lowest per-capita emissions in the partition, while Cluster 5 achieves moderate emissions in line with its high macro-efficiency (GDP per unit of energy use) and diversified supply structure.

6.4.7 Country-Level Extremes within Archetypes

Inspecting the membership of the highest-emitting archetypes provides a concrete global picture of “who emits most” within each regime. In the high-demand archetype (Cluster 1), the upper tail is dominated by Luxembourg, the United States and Canada (≈ 15 – 16 tCO₂ per capita), reflecting affluent, consumption-intensive economies where very high energy use translates into large per-capita footprints. In the fossil-tilted, high-emission archetype (Cluster 6), the highest values are instead associated with industrialized systems such as Korea, Czechia and Belgium, where emissions remain elevated due to a more carbon-intensive supply structure combined with above-average demand. Importantly, large emitters in absolute terms, most notably China and India, do not appear among the extreme “top per-capita” countries in this comparison. This is not a contradiction: the outcome metric is per-capita CO₂, so very large-population economies can account for a substantial share of global total emissions while still ranking lower on a per-person basis. Accordingly, the per-capita upper tail is dominated by high-income, high-demand economies, whereas large-population countries primarily drive global totals through scale rather than exceptionally high per-capita footprints.

6.4.8 Emissions Dynamics over Time

Finally, emissions dynamics over 2015–2021 highlight heterogeneous decarbonization paths across archetypes (**Table RQ4-2**). Cluster 1 records the steepest decline (slope ≈ -0.29 tCO₂ per year; $\approx -13\%$), indicating that the highest-emitting, high-demand economies are also those reducing emissions most rapidly in the recent period. In contrast, the decline is weaker in Cluster 6 ($\approx -4\%$) and in the gas-dominated and mixed profiles, while Cluster 3 shows a modest increase ($\approx +7\%$), consistent with low-emission economies experiencing rising energy demand during development. Within high-emitting archetypes, country-level trends reveal additional dispersion, for example, Luxembourg and the United States drive the strongest reductions in Cluster 1, whereas within Cluster 6 some countries decline markedly while others remain flat or increase.

Cluster	% change
1	-13.00
6	-3.93
5	-9.53
2	-1.16
4	-2.31
3	7.13

Table RQ4-2

6.4.9 Summary of Results

Overall, this outcome-based assessment reinforces the substantive relevance of the clustering: the six archetypes are not only interpretable in terms of energy-system structure, but they also map into distinct emissions regimes. Together with RQ1–RQ3, this typological view supports the idea that mitigation priorities differ by archetype demand-side efficiency and economy-wide decarbonization in high-demand systems, versus accelerated fuel switching and clean generation deployment in fossil-tilted profiles.

ROBUSTNESS CHECKS

Robustness of the results was assessed through a series of complementary checks designed to verify that the main findings are not driven by specific modelling choices or data configurations:

- **Comparison across econometric specifications**

Results were compared across Ordinary Least Squares (OLS), linear mixed-effects models, and mixed-effects models with an AR(1) error structure. While OLS provides a baseline estimate of average relationships, mixed-effects models account for unobserved country-level heterogeneity, and the AR (1) specification captures temporal dependence in emissions. Across all specifications, the direction and relative magnitude of key coefficients remained stable.

- **Consistency between econometric and machine-learning models**

The econometric results were evaluated alongside predictions from Random Forest and XGBoost models. Despite methodological differences, machine-learning models consistently identified electricity demand, energy efficiency, fuel composition, and renewable energy as the most influential predictors, closely aligning with the mixed-effects estimates.

- **Stability of clustering outcomes**

Cluster robustness was examined by repeating the clustering analysis using alternative subsets of energy-system and economic variables. The resulting country groupings and their qualitative interpretations remained largely unchanged, indicating that the identified energy-profile archetypes are not sensitive to specific variable selections.

- **Overall robustness conclusion**

Across all robustness checks, the core findings namely the dominant roles of electricity demand, energy efficiency, fuel composition, and renewable energy deployment remained stable, supporting the reliability and generalizability of the study's conclusions.

INTERPRETATION OF DEVIATIONS AND LIMITATIONS

The fair-residuals analysis highlights systematic country-level deviations from structurally expected per-capita CO₂ emissions, indicating persistent over- or under-emission relative to observable drivers such as electricity demand, energy mix, and economic structure. These deviations reflect structural performance rather than normative assessments and suggest the influence of unobserved factors, including policy, institutional, or technological differences.

Several limitations should be acknowledged. First, the analysis is observational and does not permit causal inference. Second, the complete-case strategy required for mixed-effects estimation may introduce selection bias by excluding countries with incomplete data, potentially underrepresenting less statistically developed economies. Third, the use of national-level aggregates masks within-country heterogeneity, particularly in large or federal states.

CONCLUSION

9.1 STUDY OVERVIEW

This study investigated the structural determinants of per-capita CO₂ emissions across countries over time using an integrated framework that combines econometric modelling, machine-learning techniques, and clustering analysis. By jointly considering energy-system characteristics alongside economic and demographic factors, the analysis provides a coherent explanation of cross-country heterogeneity in carbon emission patterns.

9.2 STRUCTURAL DRIVERS OF EMISSIONS

Results from the mixed-effects AR (1) model identify electricity demand intensity and energy-system composition as the dominant drivers of per-capita emissions. High electricity consumption, urbanization, and reliance on fossil fuels particularly coal are systematically associated with higher emissions, while renewable energy deployment, structural energy efficiency, and nuclear generation exert strong mitigating effects. These relationships are economically meaningful and closely aligned with theoretical expectations.

9.3 MACHINE-LEARNING VALIDATION

Machine-learning models reinforce these findings by achieving high predictive accuracy across multiple validation schemes. The close correspondence between SHAP-based feature importance and econometric estimates provides convergent evidence that the same core variables govern both explanatory and predictive performance, highlighting the complementary role of machine learning as a validation tool.

9.4 COUNTRY-LEVEL DEVIATIONS

Beyond average relationships, the fair-residuals analysis reveals persistent country-level deviations from structurally expected emissions, indicating substantial heterogeneity unexplained by observable drivers. The absence of a systematic relationship between the magnitude and volatility of these deviations suggests that structural inefficiency and temporal instability represent distinct dimensions of emissions performance.

9.5 ENERGY-SYSTEM ARCHETYPES

Finally, the clustering analysis demonstrates that contemporary national energy systems can be summarized into a limited number of interpretable archetypes, capturing distinct combinations of electricity demand, fuel composition, and efficiency. While energy transitions are continuous and overlapping, this typological perspective complements regression-based analysis by highlighting structural similarities and contrasts across countries.

FUTURE WORK

Future research could address the limitations of this study by adopting causal inference approaches to better isolate the effects of specific policies and structural changes on emissions outcomes. Incorporating more recent post-pandemic data would help assess whether observed relationships have shifted in response to recent economic and energy shocks. Further extensions could move beyond national averages by exploiting sub-national data where available, allowing for the analysis of within-country heterogeneity.

Additional refinements may include the integration of explicit policy indicators such as carbon pricing, renewable support schemes, or efficiency standards as well as sector-specific measures of energy use and productivity. Such extensions would strengthen the policy relevance of the framework and deepen understanding of the mechanisms underlying cross-country differences in per-capita CO₂ emissions.

REFERENCES

DATA SOURCE: World Bank. (2025). World Development Indicators 2013–2025. Retrieved from <https://data.worldbank.org/>

OTHERS:

- Ahmed, M., Li, Z., & Kaushik, S. (2023). CO₂ emission trends and drivers: A data-driven analysis. ResearchGate preprint
- Dinda, S. (2004). Environmental Kuznets Curve hypothesis: A survey. *Ecological Economics*, 49(4), 431–455.
- Liu, Z., Zheng, X., Wang, S., Li, X., & Xu, G. (2022). Drivers of global carbon emission changes: A decomposition and analysis. *Frontiers in Environmental Science*, 10, 1062500.
- Mosconi, E., Colantoni, A., & Cicia, G. (2020). Revisiting the Environmental Kuznets Curve for CO₂ emissions: Spatial variability and policy implications. *Economies*, 8(4), 74.
- Salahuddin, M., Alam, K., & Ozturk, I. (2016). CO₂ emissions in G20 economies: A dynamic panel analysis of economic and energy sector drivers. *Environmental Economics*, 7(4), 32–45.
- Wang, Q., & Wang, S. (2020). Renewable energy consumption and economic growth: A panel data analysis. *Energy*, 207, 118200.

